

# Predator\_5

Learning a Proof-Search Policy from Computed Geodesics  
Logistic Regression against Random Forests over Condensed Detachment

Brian Tenneson

M.S., Applied Data Science  
btenneson2301@baypath.edu

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## Abstract

Predator\_4 ranks lemmas against a goal. It has no search: it emits an ordering and stops. Predator\_5 ranks *edges against a state* — from proof state  $p$ , which admissible one-step extension moves toward a shortest proof? — and that ordering is consumed by a search that extends states until the target appears. This is a nondeterministic proof-search policy  $\Sigma$  whose nondeterminism has been ordered by a fitted model.

Three consequences follow, and each is load-bearing. First, the negatives change: a policy's negatives must be the other edges *applicable at that state*, not lemmas sampled from elsewhere in the library. Second, the labels must be computed rather than read: a human-written proof is an upper bound on the shortest proof, not the shortest proof, so training on one teaches what humans wrote. Predator\_5 runs breadth-first search first, obtaining true distances, and marks every edge lying on some shortest path. Third, there is a completeness theorem to lose. Ordering  $\Sigma(p, m)$  preserves proof-covering; truncating it does not.

On held-out targets in an implicational fragment, the learned policy reduces mean node expansions from 24.4 to 13.8, a factor of 1.77, while returning a shortest proof on every target it solved. Two findings run against expectation. Logistic regression *beat* random forests (13.8 against 17.8), indicating the policy signal in these features is close to linear. And pruning — which surrenders proof-covering and hence the Branch-Covering Theorem — was not merely unsound but empirically worse: forest-with-pruning solved 89% of targets against 100% for every complete policy, and expanded 30.3 nodes on average, more than unguided breadth-first search. Pruning here bought nothing and cost a theorem.

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## 1 What changed from Predator\_4

Predator\_4 answers: given a goal, which of set.mm’s 43,900 statements does its proof cite? It is a ranking over lemmas, trained on the premise lists written in the file, and it contains no inference machinery of any kind.

Predator\_5 answers a different question: from proof state  $p$ , which admissible one-step extension moves toward a shortest proof? Three differences follow.

### 1.1 The negatives are different

Predator\_4 samples non-premises from the whole library. A policy cannot do this. Its negatives must be the other edges *applicable at that state*, because those are the alternatives it actually has to rank against. Ranking an edge against a lemma drawn from an unrelated part of the search answers a question no search ever asks.

### 1.2 The labels are computed, not read

A proof recorded in a library is an *upper bound* on the shortest proof. The set.mm repository’s history is full of shorter proofs replacing longer ones. Train on those proofs and the model learns what humans wrote, which is correlated with, but not equal to, what lies on a geodesic.

Worse, a recorded proof is the path taken and never the frontier rejected, so it supplies no state-level negatives at all.

Predator\_5 therefore computes its own labels. By the shortest-path principle, breadth-first search on the novelty state graph reaches a target at its true graph distance. A backward pass over *all* minimal-distance predecessors — all of them, not one arbitrarily chosen path — marks every edge lying on some shortest path. Those edges are the positives; every other edge offered at a state on some geodesic is a negative. The labels are correct by construction.

The price is that the fragment must be small enough for breadth-first search to terminate. Section 7 discusses what this rules out.

### 1.3 There is a theorem to lose

The Branch-Covering Theorem requires  $\Sigma$  to be *proof-covering*: for every finite proof  $\pi = \langle \pi(1), \dots, \pi(n) \rangle$  from  $\Gamma$  there is a branch  $b = \langle p_1, p_2, \dots \rangle$  with  $\{\pi(1), \dots, \pi(j)\} \subseteq p_j$  for every  $j \leq n$ . Under that hypothesis, if  $\Gamma \vdash_F \varphi$  then some target-search branch reaches  $\varphi$  by stage  $\|\varphi\|_F(\Gamma)$ .

Two ways to use a learned score, with different standing:

**Proposition 1.1** (Reordering preserves proof-covering). *Let  $\Sigma_{\text{full}}$  be the policy returning every admissible one-step extension at  $p$ , and let  $\Sigma_\theta$  return the same set permuted by a scoring function  $\theta$ . If  $\Sigma_{\text{full}}$  is proof-covering then so is  $\Sigma_\theta$ .*

*Proof.* Proof-covering quantifies over the branches available from  $\Gamma$ , and a branch is determined by which extensions are *present* in each  $\Sigma(p_m, m)$ , not by the order in which they are listed. Since  $\Sigma_\theta(p, m) = \Sigma_{\text{full}}(p, m)$  as sets, the two policies generate exactly the same set of branches. Any branch shadowing  $\pi$  under  $\Sigma_{\text{full}}$  therefore shadows  $\pi$  under  $\Sigma_\theta$ .  $\square$

**Proposition 1.2** (Truncation destroys it). *Let  $\Sigma_{\theta, k}$  return only the top  $k$  extensions by  $\theta$ . Then  $\Sigma_{\theta, k}$  need not be proof-covering, and the Branch-Covering Theorem’s hypothesis fails for it.*

*Proof.* It suffices to exhibit a state at which the unique extension continuing a shortest proof is scored outside the top  $k$ . Since  $\theta$  is fitted and no constraint ties its ordering to geodesic membership, nothing prevents this, and Section 5 reports it occurring: pruned policies failed on targets that every complete policy solved. Once the extension is deleted, no branch of the truncated search contains  $\pi(j)$  at stage  $j$ , so the covering condition fails at that  $\pi$ .  $\square$

*Remark 1.3.* Consequently a speedup obtained by reordering and a speedup obtained by pruning are not the same kind of object. Reporting them in one column without saying so is the single easiest way to mislead a reader here, which is why every results row in Predator\_5’s output carries its guarantee alongside its number.

## 2 The search

### 2.1 The interpolation knob

Search priority is

$$f(q) = \text{depth}(q) - \lambda \cdot \text{score}(e \rightarrow q).$$

The parameter  $\lambda$  is not free. It interpolates between two objects already in the theory:

- $\lambda = 0$  gives  $f = \text{depth}$ , which is breadth-first search. The first proof found is shortest and the search is complete.
- $\lambda \rightarrow \infty$  gives  $f = -\text{score}$ : greedy descent, with no guarantee of any kind.

Between the endpoints, and under reordering, the search remains complete — every state is eventually popped — but the first proof found need not be shortest, because a fitted score is not an admissible heuristic and essentially never is. A length reported under a guided row is therefore an *upper bound* on  $\|\varphi\|_F(\Gamma)$ ; only the breadth-first row reports the geodesic itself.

*Remark 2.1* (The depth term is not a refinement). An earlier version of this program ordered by learned score alone. That is pure greedy best-first: nothing in the priority decreases with depth, so the search chases high-scoring states downward and never returns to the breadth that breadth-first search gets for free. It *lost* a target that breadth-first search reaches in twenty expansions, inside a budget of four hundred. The depth term is the thing that makes the search work at all.

## 2.2 The fragment

Condensed detachment over implicational formulas. From a major premise  $A \rightarrow B$  and a minor premise  $C$ , rename apart, unify  $A$  with  $C$ , and return  $B$  under the most general unifier. Unification carries the occurs check: it is not optional, since without it detachment produces cyclic terms and the novelty state graph stops being acyclic.

Take  $\Gamma = \{K, S, W\}$  with

$$K = a \rightarrow (b \rightarrow a), \quad S = (a \rightarrow (b \rightarrow c)) \rightarrow ((a \rightarrow b) \rightarrow (a \rightarrow c)), \quad W = ((a \rightarrow b) \rightarrow a) \rightarrow a.$$

Then  $\Sigma(p, m)$  is every novel  $D(x, y)$  for ordered pairs  $x, y \in p$ , which is finite and enumerable. That property is exactly what `set.mm` lacks, and exactly why `Predator_2` abandoned forward search for premise selection.

## 2.3 Targets are harvested, not hand-picked

One bounded breadth-first sweep from  $\Gamma$  reaches many states. A formula first appearing at layer  $d$  has shortest-proof length exactly  $d$ . So a single sweep prices dozens of targets at once, and they can be split into training and held-out sets.

This matters. The first draft of this experiment hand-picked four named theorems, trained on two of them, and evaluated on the same two. It reported a  $3.3\times$  speedup. That number measured memorisation and nothing else. A sweep to depth 4 costs 0.1 seconds and yields 63 targets with exact geodesics, which is enough for a real split.

## 3 Features

Twelve features of the (state, edge) pair. Every one varies across the edges offered at a single state; a feature constant across them cannot change their order, and the fit will still assign it a weight, which is worse than useless because it then looks informative in the printout.

| Feature                               | Meaning                                 |
|---------------------------------------|-----------------------------------------|
| conclusion size                       | $ \text{concl} /10$                     |
| conclusion size / max in state        | relative to the largest formula present |
| conclusion novel-variable ratio       | variables not occurring in the target   |
| major size, minor size                | sizes of the two premises               |
| major in $\Gamma$ , minor in $\Gamma$ | is the premise an axiom                 |
| unifier growth                        | $ \text{concl} / \text{major} $         |
| state size                            | $ p /10$                                |
| top-shape matches target              | same outermost shape as the target      |
| subterm overlap with target           | Jaccard over subterm sets               |
| conclusion variable count             | $ \text{vars} /5$                       |

Both rankers are fitted by pairwise ranking loss on difference vectors  $x_p - x_n$ , with the negated copies labelled 0 for antisymmetry. Without those copies the fit can satisfy the objective by inflating every score rather than by ordering anything.

## 4 Experimental protocol

1. Sweep breadth-first from  $\Gamma$  to depth 4. Record every state's distance and *all* minimal-distance predecessors.

2. Harvest every formula reached, with its exact geodesic. Discard targets at distance 1: every policy finds them immediately.
3. Split targets into training and held-out sets by seed.
4. For each training target, mark on-geodesic edges and form (positive, negative) pairs *within each state*.
5. Fit logistic and forest rankers on those pairs.
6. Evaluate on held-out targets only: mean node expansions, solve rate, and how often the returned length equals the geodesic.

Step 6 is the one that matters. The held-out targets were never seen during fitting, so the numbers below are generalisation and not recall.

## 5 Results

Depth 4, 19 held-out targets, expansion budget 120,  $\lambda = 0.5$ , prune width  $k = 4$ .

| Policy                          | Solved     | Mean exp.   | Optimal | Guarantee          |
|---------------------------------|------------|-------------|---------|--------------------|
| Breadth-first ( $\lambda = 0$ ) | 100%       | 24.4        | 100%    | geodesic, complete |
| <b>Logistic, reorder</b>        | 100%       | <b>13.8</b> | 100%    | complete           |
| Logistic, prune $k=4$           | 100%       | 17.7        | 95%     | <b>incomplete</b>  |
| Forest, reorder                 | 100%       | 17.8        | 100%    | complete           |
| Forest, prune $k=4$             | <b>89%</b> | <b>30.3</b> | 94%     | <b>incomplete</b>  |

### 5.1 Finding 1: the linear model wins

Logistic regression reached targets in 13.8 expansions against the forest’s 17.8. A forest can represent interactions and non-monotone effects that a linear score cannot, so the expectation was the reverse.

The honest reading is that the policy signal in these twelve features is close to linear, and the forest’s extra capacity is spent fitting noise in a training set of 707 pairs. This is evidence about the *task*, not evidence that the forest is broken. Feature importances support it: one feature, subterm overlap with the target, carries an importance of 0.52 with the remaining eleven splitting the rest.

There is also a cost asymmetry not visible in the table. The forest calls `predict_proba` once per state expansion, and at 300 trees that dominates wall-clock time even when it expands fewer nodes. The runs above use 30–60 trees for this reason.

### 5.2 Finding 2: pruning cost a theorem and bought nothing

This is the result worth keeping.

Pruning to the top  $k$  forfeits proof-covering (Proposition 1.2) and therefore the Branch-Covering Theorem. The usual justification is that the speed is worth the guarantee. Here there was no speed:

- Logistic with pruning expanded *more* nodes than logistic with reordering (17.7 against 13.8) and dropped from 100% to 95% optimality.

- Forest with pruning expanded 30.3 nodes — *more than unguided breadth-first search at 24.4* — and solved only 89% of targets, failing on two that every complete policy solved.

The mechanism is straightforward. Deleting extensions does not delete the work; it displaces it. When the deleted edge was the one continuing a shortest proof, the search must reach the target the long way round, and the extra depth costs more expansions than the narrower frontier saved.

*Remark 5.1.* Proposition 1.2 predicts the failures. It does not predict that pruning would also be slower — that is empirical, and it may not survive a larger fragment or a wider  $k$ . It is reported because it is what happened.

## 6 Using it

### 6.1 Install

```
# logistic ranker only -- no dependencies at all
python predator5.py doctor

# for --model forest
python -m pip install numpy scikit-learn
```

### 6.2 Harvest: see the ground truth before training on it

```
python predator5.py harvest --depth 4 --edge-cap 15 --max-size 14
```

Prints how many states the sweep reached, how many targets it priced, the distribution of geodesic lengths, and the held-out target list. Run this first: if the sweep finds too few targets at depth  $\geq 2$ , every later number rests on a handful of instances.

### 6.3 Train one model

```
# logistic
python predator5.py train --model logistic --depth 4 --seed 0

# random forest
python predator5.py train --model forest --depth 4 --seed 0 \
    --n-estimators 300 --max-depth 12 --min-samples-leaf 4
```

Prints weights for the logistic ranker, Gini importances for the forest. A weight whose sign contradicts intuition is worth reading closely: it is often a feature that varies little within a state and is absorbing effect from a correlated one.

### 6.4 Compare, which is the command that matters

```
python predator5.py compare --depth 4 --edge-cap 12 --max-size 14 \
    --budget 150 --lam 0.5 -k 4 --test-frac 0.35 \
    --n-estimators 60 --max-depth 8 --out results.json
```

Fits both rankers on the training targets, then evaluates both — in reorder and prune mode — against breadth-first search on the held-out targets. Writes JSON with `--out`.

## 6.5 Sweeping the interpolation knob

```
for L in 0 0.25 0.5 1 2 4 8; do
  echo "=== lambda = $L ==="
  python predator5.py compare --depth 4 --lam $L --budget 150 \
    --n-estimators 40 --out "lam_$L.json" | grep -A8 "HELD-OUT"
done
```

At  $\lambda = 0$  every row must reduce to breadth-first; if it does not, the depth term is wired wrong. As  $\lambda$  grows, expansions should fall and then rise as the search starts diving. The optimality column should decay monotonically.

## 6.6 Checking that the split is real

```
for S in 0 1 2 3 4; do
  python predator5.py compare --depth 4 --seed $S --budget 150 \
    --n-estimators 40 | grep "logistic reorder"
done
```

Different seeds give different train/test splits. If the speedup swings wildly across seeds, the held-out set is too small to support the claim — which at 19 targets is a live concern, and the honest reason to widen the sweep before quoting any figure from this paper as settled.

## 7 What this does not do

### 7.1 It does not run on set.mm

The method requires breadth-first search to terminate, because that is where the labels come from. In set.mm the one-step extension set is not enumerable in practice: re-deriving a proof needs the full substitution machinery over a vocabulary of thousands of symbols. This is the same wall Predator\_2 hit, and it is the reason Predator\_2 became a premise selector instead of a searcher.

Predator\_4 and Predator\_5 are therefore complementary rather than sequential. Predator\_4 scales to set.mm and cannot search. Predator\_5 searches and does not scale to set.mm. Bridging them means either enumerating extensions in set.mm — which is the hard open engineering problem — or learning a policy on small fragments and testing whether it transfers, which the present features almost certainly do not, since they are stated in terms of condensed detachment.

### 7.2 The hyperreal target

A recurring request has been to point this apparatus at

$$\text{Inf} \approx \mathbb{R},$$

the claim that the infinitesimals of the ultrapower  $\mathbb{R}^{\mathbb{N}}/F$  are equipollent to  $\mathbb{R}$ . Three obstacles, in increasing order of seriousness.

*It is not expressible in set.mm.* There is no ultrapower construction, no  $^*\mathbb{R}$ , and no class of infinitesimals. The token  $\text{RR}^*$  denotes the extended reals  $\mathbb{R} \cup \{\pm\infty\}$ , a different object. A prover cannot attempt a formula that is not well-formed in its language, so this is prior to any question of search or training.

*Predator\_4 cannot prove anything.* It ranks and stops. This is not a limitation of its training data.

*Predator\_5 can search, but not there.* Its rule is condensed detachment over implicational formulas, not ZFC.

The statement is true and its proof is short:  $r \mapsto r\varepsilon$  for a fixed positive infinitesimal  $\varepsilon$  injects  $\mathbb{R}$  into  $\text{Inf}$ ;  $\text{Inf} \subseteq {}^*\mathbb{R}$  and  $|{}^*\mathbb{R}| \leq |\mathbb{R}^{\mathbb{N}}| = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$ ; apply Schröder–Bernstein. The endgame lemmas exist in `set.mm` today (`sbth`, `mapdom`, `rpnnen`). What does not exist is the construction: a nonprincipal ultrafilter on  $\mathbb{N}$ , the quotient, its order, and the definition of  $\text{Inf}$ . `set.mm` has filters and ultrafilters (`df-fil`, `df-ufil`) and choice (`ax-ac`), so the foundation is present; assembling it is a formalisation project of several hundred statements.

Once those definitions exist and sit *after* everything a model trained on, the out-of-distribution experiment becomes well-posed — and it is a good experiment. It is simply downstream of the formalisation, not a substitute for it.

## 8 Conclusion

A search policy trained on computed geodesics reduced held-out node expansions by  $1.77\times$  while returning a shortest proof every time it succeeded. The linear model beat the forest, which says the signal in these features is close to linear. Pruning the policy surrendered proof-covering and, in this fragment, purchased nothing: the pruned forest was slower than unguided breadth-first search and solved fewer targets.

The methodological point outlives the numbers. Labels for a search policy should be computed, not read from a library of human proofs, because those proofs are upper bounds on the geodesic and record no rejected frontier. Where breadth-first search can be run, it supplies both the ground truth and the benchmark, and the resulting comparison is honest in a way that training on recorded proofs cannot be.

The held-out set is 19 targets. That is enough to see an effect and not enough to size it. Widening the sweep is the next task, and the seed-sweep in Section 6.6 is how to find out whether anything here is stable.